

“PLANT EXTRACT DEPENDENCE SYNTHESIS OF NICKEL OXIDE”

A dissertation

Submitted in partial fulfilment of the requirements of
the degree of

Master of Science

in Physics

by:

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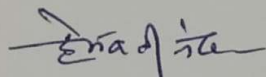


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CERTIFICATE

Certified that the work incorporated in the dissertation entitled "**plant extract dependence synthesis of nickel oxide**" submitted by **Prachi Alariya** was carried out by the candidate, under my supervision of **Dr. Shubhra Kala**. The work presented here or any part of it has not been included in any other dissertation submitted previously for the award of any degree or diploma from any other university or institution.

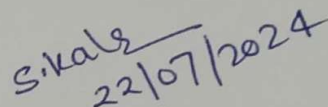


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I hereby certify that the work which is being presented in the dissertation entitled “**Plant extract dependence synthesis of nickel oxide**” in partial fulfilment of the requirements for the award of the Degree of Master of Science and submitted to the Department of Physics at H.N.B.Garhwal University (A Central University) is an authentic record of my work carried out during a period from the year 2023-2024 under the supervisor of **Dr. Shubhra Kala, Assistant Professor** Department of Physics, H.N.B. Garhwal University (A Central University). The matter presented in the dissertation has not been submitted by me for the award of any other degree of this or any other institute.

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ACKNOWLEDGEMENTS

First, I would like to express my sincere gratitude to my supervisor “**Dr. Shubhra Kala**” for giving me an opportunity to work on the topic titled “**Plant Extract dependence synthesis of Nickel Oxide**” and consistently guiding and supporting me throughout the dissertation work.

I am very thankful to **Prof. Hemwati Nandan** (Head of Department of physics) for providing all the facilities during the Dissertation work. I am thankful to my teachers Prof. S.C. Bhatt, Dr. Alok Sagar Gautam, Dr. Meera Rawat, Dr. Manish Uniyal (Faculty member in the Department of Physics) and my classmates for their encouragement and co-operation throughout the Dissertation work.

I am also very thankful to the research scholars **Mr. Shailendra Chamola** for precious support.

I am also very thankful to my batch mates and especially **Gaurav Singh, Taku karo, Vandana Rawat** to whom I work with for my dissertation work.

Thank You

Prachi Alariya

ABSTRACT

The green method used for synthesis of nanoparticles is an efficient and ecofriendly method. Different biological sources can be used in this process for formation of various nanoparticles based on their applications. This study investigates the synthesis of nickel oxide nanoparticles by green method using different plant extract. For this purpose, we have chosen one yellow elder flower and one Indian bay leaf, The main goal is to investigate the formation of nanoparticles by these two kind of extract. Here we have taken Ni precursor to observe this dependence. The formed nickel oxide nanoparticles were characterized using UV-Vis spectroscopy, X-ray diffraction (XRD) and scanning electron microscope (SEM) to analyze their optical structural and morphological properties, respectively.

Keywords: nickel oxide, nanoparticles, Green synthesis, X-rays diffraction, UV-Visible spectroscopy, SEM(scanning electron microscopy).

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CHAPTER-1 INTRODUCTION

1.1 Nanoparticles

The term 'nanotechnology' describes the atomic, molecular, and macromolecular size research and technology development in this technology a very tiny particles whose diameter is in between 1- 100nm, particles in this range called "nanoparticles", this Greek word 'nano,' which meaning dwarf or small, is where the name "nanoparticle" originated[1]. This field of science that research this stuff called nanoscience and it's produced loads of new materials that have useful properties. Nowadays nanotechnology is widely use in different areas because of their small size and different physical and chemical properties, one of the key features of nanoparticle is that they have large surface area to volume ratio. When compared to big particles or atoms, nanoparticles have a surface to volume ratio that is 35–45% higher. This particular extrinsic characteristic of the nanoparticle's surface area contributes to its high value and influences other intrinsic characteristics like the size-dependent strong surface reactivity. nanomaterials are also classified on the bases of their dimensions like fullerenes, quantum dot, nanotubes etc. Applications for nanoscale materials can be found in many different fields, including biological, pharmaceutical, cosmetic, energy, environmental, magnetic and optoelectronic, electronic etc[2]. There are two types of synthesis approaches for nanoparticles: top-down and bottom-up. Top-down methodologies often involve the use of externally controlled tools, such as lithography methods, laser beams, and mechanical techniques, to cut, mill, and shape materials into the desired nanoscale structures. Bottom-up methods, like the previously mentioned chemical synthesis, take advantage of the molecules' chemical characteristics to drive self-assembly into nanoparticles. Compared to top-down approaches, bottom-up methods are more economical and can yield considerably smaller nans. The synthesis of core-shell structured nanoparticles is done using both techniques. The quality of the environment's air, water, and soil might all be greatly enhanced by nanotechnology. It can enhance pollution detection and sensing and support the creation of innovative cleanup solutions[3]. Comprehending the dynamic processes of nanoparticle production and growth (in the combustion system, for example) enables the creation of effective techniques to minimize the creation of pollutants initially and to lower their emissions. Although nanotechnology has

the potential to improve environmental quality, there are concerns that it can also lead to a new class of environmental hazards. Therefore, we required a cheap and environmentally ways to synthesized NP's.

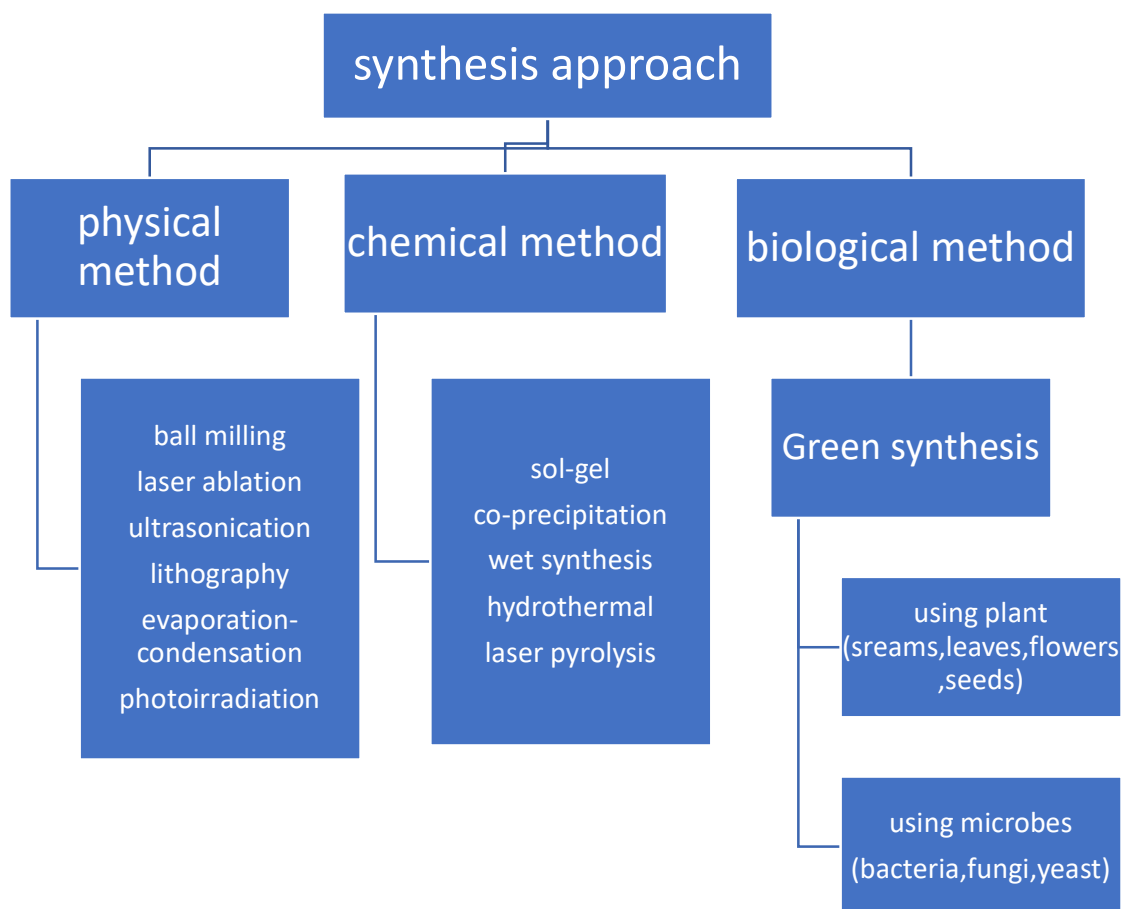


Fig1: Flowchart showing various method of synthesis of nanoparticles

1.2 Green synthesis

Biological methods include the use of bacteria, fungi, yeast and plant extract. These methods are also called green synthesis or green method as they are eco-friendly method. Green synthesis techniques make use of moderately pollutant free chemicals to synthesis NPs and embrace the use of benign solvents such as water, natural extracts. Green chemistry seeks to reduce pollution at source. Green method is more effective in producing NPs whereas, the environmental impact of producing nanoparticle (NPs) through chemical and physical methods has been increased because of their hazardous effects. Use of more chemicals, which are hazardous and poisonous to the environment and human health, may increase particle reactivity and toxicity and have unintended negative health impacts. The potential for green synthesis strategies to lower NP toxicity makes them very appealing. As a result, the usage of vitamins, amino acids, and plant extracts is becoming increasingly common in modern times. Plants are easily available and easy to use and eco- friendly too. Hence, we can use any part of plant such as Flower, steam, leaves, roots etc. for plant extract. This extract is used as reducing and capping agents for the synthesis of NPs because plants contain many photochemical such as amides, ketone, flavones, aldehydes, carboxylic acid, terpenoid and phenol these are capable of reducing the particles in nano range. The basic properties of these nanoparticles have been noted for applications in molecular sensing, optical imaging, biomedical diagnostics, antimicrobials, catalysis, and biological system labelling. Using the flower-mediated green approach, numerous plants have been repurposed for synthesis. Various flowers including, *Zephyranthes rosea*, *Sesbania grandiflora*, *Salvia hispanica*, *Allium cepa*, *Arabic gum*, *Rhamnus virgata*, *Populus ciliata*, etc. have been used for the synthesis of silver, gold and other NPs. Some of reported nanoparticle that are synthesis using plant extract are listed below in table 1.

Table 1. Various plant involves in synthesis of various NPs [6- 20].

S. No.	Plant	Plant material used	NP synthesized	Size of NPs (nm)	Shape
1.	<i>Zephyranthes rosea</i>	Flowers	silver	27.148	-
2.	<i>Salvia hispanica</i> L. (chia)	seeds	silver	33	cubic
3.	Arabic gum	gum	silver	39-62	cubic

4.	<i>Rhamnus virgata</i>	leaves	nickel	24	cubic
5.	<i>Azadirachta indica</i> (L.)	leaves	Zinc	18	-
6.	<i>Deverra tortuosa</i>	aqueous	Zinc	15.41	-
7.	<i>Citrus medica</i>	leaves	silver	40-220	spherical
8.	<i>Nyctanthes arbor-tristis</i>	flower	zinc	12-13	hexagonal
9.	<i>Aerva lanata</i>	flower	silver	5-15	-
10.	<i>Malva sylvestris</i>	flower	silver	20-40	-
11.	<i>Abelmoschus esculentus</i>	seeds	gold	45-75	cubic
12.	<i>Artocarpus heterophyllus</i> Lam.	seeds	silver	10.78	irregular
13.	<i>Pistacia atlantica</i>	seed	silver	27	cubic
14.	<i>Astragalus tribuloides</i>	root	silver	34.2 ± 8.0 nm.	spherical
15.	<i>Lepidium draba</i>	root	silver	24	-
16.	<i>Garcinia mangostana</i>	peels	gold	32.96 ± 5.25	spherical
17.	<i>Citrus maxima</i>	peels	iron	10-100	irregular
18.	banana	peel	silver	23.7	spherical
19.	<i>Citrus limon</i>	leaves	silver	<50	spherical
20.	<i>Panax ginseng</i>	root	gold	10-40 and 10-30	spherical

1.3 Nickel oxide (NiO)

Among the various nanomaterials, metal oxides with NiO inclusive have attracted increasing technological and industrial interest. This interest has mainly to do with their properties optical,

magnetic, electrical, and catalytic properties associated with general characteristics such as mechanical hardness, thermal stability or chemical passivity. A variety of metals, including silver, gold, platinum, magnesium, iron oxide, calcium oxide, and zinc oxide, have been produced up to this point. Among these prepared NPs, scientists in a variety of fields have been captivated by nickel oxide NPs (NiONPs) because of their simple synthesis, small size, wide bandgap, and semiconductor characteristics. Numerous successful applications of NiONPs in batteries, sensors, semiconductors, magnetic material photocatalysis, and catalytic analysis have been analysis. Three primary sources are used in green synthesis: autotrophs (plants, flowers, and algae), bacteria, and fungi. Due to its simplicity, accessibility, eco-friendliness, and non-toxicity, flower-mediated green synthesis is gaining importance in the biomedical area. It also eliminates the need for reducing agents and energy from the outside environment. The significant impact of nickel on the oxidation resistance of high temperature alloys and the model role of nickel oxide in research on the semiconducting properties of transition metal oxides are the primary causes of the increased interest in the physicochemical aspects of the nickel–oxygen system. Nickel oxide (NiO) has been reported to exhibit a rock-salt structure and be a metal deficit p-type semiconductor, with cation vacancies and electron holes being the main defects. Because of its many uses in chemical sensing, batteries, photocatalysis, and electrochromic technology, nickel oxide (NiO) is a highly desirable semiconductor. Nio is also a desirable material for use in the production of nickel cermet, which is used in the anode layer of solid oxide fuel cells, in the cathodes of lithium nickel oxide for lithium-ion micro batteries, in textiles, plastics, and electrochromic coatings, as well as in the production of nanowires, nanofibers, and particular alloy and catalyst applications. Additionally, it finds application as an antiferromagnetic layer and catalyst in lightweight aircraft structural components, active optical filters, cathode materials for alkaline batteries, and materials for temperature or gas sensors as formaldehyde, CO, and H sensors. Through the process of thermal evaporation, nickel oxide thin films were produced, and the required characteristics were noted, enabling their application in transparent heat mirror multilayer structures that conserve energy. Their directed crystalline formation and smooth, dense surfaces with sub-nano-meter surface roughness were indicators of this. NiO is utilized in the production of ferrites (nickel zinc ferrite), glazes, glasses, and electrical ceramics like thermistors and varistors. In optoelectronic devices, sheets of nickel oxide doped with gold can be employed as transparent electrodes.

Among metal oxides, nickel oxide is a well-known example that can be produced by thermally breaking down freshly made nickel hydroxide by sol gel route at 300°C (572°F). It has a melting point of 1955°C and a density of 6.67g/cm³.

Table.2 Reported synthesis of NiO nanoparticles by various plants^[21-23]

S.No.	Plant	Part of plant used	Size NPs (nm)	Morphology	Application
1.	Populus ciliata	leaves	44	cubic	used as an ingredient in herbal cough medicines.
2.	Acacia nilotica	leaf	>16	cubic	primary applications in oral & dental hygiene, burn injuries and in the management of skin diseases
3.	Avicennia Marina	leaves	30-100	-	Used for petroleum ether, chloroform, ethyl acetate and ethanol against Escherichia coli
4.	Calotropis gigantea	leave	20-40	-	Crushed leaves can be applied to treat burns, headaches, and rheumatic pains
5.	Zea mays lea silk	leaf	10-20	spherical	Some studies indicate that compounds in leaves may have anti-cancer
6.	Solanum trilobatum	leaf	23.21	cubic	used to treat respiratory issues such as chest congestion, sinusitis, bronchial asthma, and tuberculosis

7.	Rheum turkestanicum	root	12-15	Rhombohedral	effective on various diseases such as diabetes, hypertension, cancer, and fatty liver
8.	Geranium wallichianum	leaves	~21	cubic	treatment of rheumatism, leucorrhoeas, arthritis,
9.	Aloe barbadensis miller	leaves	25-55	spherical	antioxidant properties that benefit the skin and hair. It also protects the skin from harmful environmental factors
10.	Avicennia Marina	leaf	30-100	-	Use as infection Prevention, Fever Reduction, Water Purification
11.	Eclipta alba	leave	3-9	Spherical	Hair Care, Liver Health, Cold and Cough,
12.	Moringa Oleifera	leave	305.46-400	cubic	vitamin C, calcium, protein, iron, Skin care, food preservative
13.	Arundo donaxi	leave	41.13	-	Use Leaves of Arundo donax have been used traditionally as a diuretic to promote urine production
14.	Oak	Fruit	35	shperical	Oak timber is a strong and hard wood with many uses, such as for furniture, floors, building frames, and veneers
15.	Nephelium lappaceum L.	peel	104	cubic	use in food, pharmaceutical and cosmetic industries
16.	limonia acidissima	fruit	10-21	cubic	used in India as a liver and cardiac tonic, effective treatment for hiccough

17.	Punica granatum	leaves	10-30	spherical	antioxidant, anti-inflammatory, antimetabolic, anticancer
18.	Parthenium hysterophorus	leaves	24-84	Spherical and hexagonal	Leaves are sometimes used to alleviate pain in traditional medicine
19.	Annona muricata L.	Fruit peel	20-90	spherical	a wide use in human diseases including inflammatory conditions, neuralgia, diabetes, cystitis
20.	stevia	leave	50	rectangular	used to sweeten beverages, desserts
21.	Ceropegia candelabrum	leaves	12-35	hexagonal	The leaves are used in traditional medicine to treat wounds and cuts.
22.	Aspalathus Linearis	leaves	$\sim 31.8 \pm 5$ nm	Quasi spherical	fermented leaves of the Aspalathus linearis shrub
23.	Euphorbia jatropha	stem	15	hexagonal	Wound Healing, Pain Relief, Soil Improvement
24.	Physalis angulata	leaf	64.13	cubic	used in traditional medicine for a long time to treat a variety of ailments,
25.	Physalis angulata	leaf	416	cubic	treatment of inflammatory illness
26.	Jacaranda mimosifolia	flower	2-4	Hexagonal wurtzite	seed in traditional medicine to treat wounds, rheumatism, colds, neuralgia, and varicose veins, skincare
27.	Ficus carica	leaf	414	cubic	used to treat a variety of health issues, including diabetes, high cholesterol, skin conditions, gastrointestinal issues

1.4 Yellow elder (Tecoma stans)

Yellow elder (scientific name *Tecoma stans*) Yellow trumpetbush, yellow bells, yellow elder, and ginger Thomas are some common names. The official flower of the US Virgin Islands and the floral symbol of The Bahamas is *tecoma stans*. It is commonly found in northern India, including along river banks and in gardens. Growing in any well-drained soil and receiving full light, because yellow elder can only be sustained by rain, it is a great choice for low-maintenance, naturalized landscapes. It works well in parking lot medians and islands, and it would also make a lovely patio tree. There may be a minor litter issue as a result of the falling fruit[26].

These plants are easily propagated from cuttings as well as from seeds. Within two years, seedlings can be easily transplanted and will begin to blossom. Use of these plants components to treat a variety of illnesses and ailments. Hyperglycemia, gastrointestinal issues, kidney issues, jaundice, skin infections, toothaches, headaches, joint pains, sore eyes, and heart pain are a few of them. It is also one of the most widely used remedies for rat, snake, and scorpion bites. Pharmacological research has demonstrated the following benefits: hypoglycemic, hypolipidemic, immunomodulatory, anticancer, antioxidant, analgesic, antibacterial, antispasmodic, wound healing, and hepatoprotective, this plant is also useful as a lubricant, cosmetic, and flavouring component and in perfumery.



Fig2: yellow elder (tecoma stans)

In this dissertation work yellow elder flower extract was used as a reducing agent for the synthesis of NiO NPs. The phytochemicals present in yellow elder flowers, such as flavonoids and phenolic compounds, are believed to facilitate the reduction of silver ions to silver nanoparticles when used in appropriate synthesis methods[27]. The aim of this study was to report the green approach for the synthesis of NiO NPs using yellow elder flower extract.

1.5 Indian bay leaf (*Cinnamomum tamala*)

Indian bay leaf, or *Cinnamomum tamala*, is a tree of the Lauraceae family that is native to Bangladesh, India, Nepal, Bhutan, and China. It is also referred to by the names tejpat, tejapatta, Malabar leaf, Indian bark, Indian cassia, and malabathrum. Its maximum height is 20 meters, or 66 feet. Indian bay leaf is found in the wild Himalayan valleys of Uttarakhand at altitudes ranging from 500–2,400 meters. This plant's dried leaves are a popular spice for flavouring in Indian homes. It contains volatile, active substances including mono- and sesquiterpenes. Among these are geraniol, linalool, phellandrene, and pinene. The monoterpene linalool, which is abundant in bay leaves, has some influence on the central nervous system. It has a strong sedative effect that is dose-dependent and possesses hypnotic, anticonvulsant, and hypothermic qualities. A popular spice in homes is bay leaf, which is used to flavour a variety of dishes. It is also employed as a cholagogue, stimulant, carminative, and diaphoretic due to its numerous therapeutic qualities. Its essential oils are used as aroma ingredients in soaps, lotions, creams, and perfumes. They also have fungicidal, insecticidal, and bactericidal qualities. bay leaves are reported to contain 0.3–3.1% of volatile oil composed mainly of cineole, α -pinene, linalool, α -terpineol acetate, α -terpineol, β -pinene, sabinene, limonene, methyl eugenol, eugenol, p -cymene, camphene, dehydro-1,8-cineole, phenyl-hydrazine, piperidine, and geraniol[28].



Fig3: Indian bay leaf (*Cinnamomum tamala*)

CHAPTER-2 EXPERIMENTAL DETAILS

2.1 Preparation of yellow elder flower solution

For sample1, Fresh flower of yellow elder was collected from Hemwati Nandan Bahuguna Garhwal university, Chauras campus, Srinagar. They were properly washed with tap water (2-3times) at first, and then each one rinsed separately with DI water (2 times). The flowers were then left to dry at room temperature for a week in the dark. When the flowers dried completely, we took 5gm of yellow elder, crushed them, after that, we added those crushed flowers to a beaker filled with 100 ml of DI water, and then heated the mixture at 60°C for 60 minutes on magnetic plate heater.

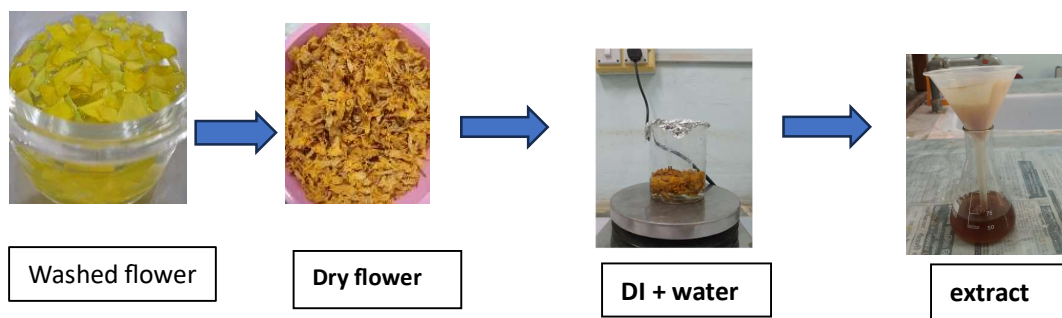


Fig4: yellow elder flower extract

After allowing the mixture to cool, it was filtered twice through Whatman filter paper to eliminate any remaining particles, the filtrate that was left behind was the necessary extract. The extract was then stored at a cool place for further experimental.

For preparation of nickel nitrate solution, we took 2.32gram of nickel nitrate in 40ml of DI water and stirrer it for 5minute to prepare a nickel nitrate solution.

2.2 Preparation of Indian bay leaf

For sample 2, Indian bay leaf was collected from local area of Srinagar. First, they were thoroughly with tap water (2-3times), and then each leaf was washed with DI water (2 times). After this the leaves were left at room temperature in isolation to dry for 7 days. The extract

was prepared by adding 5.000 gm of crushed leaves in 100 ml of double distilled water. and then heated the mixture at 60°C for 30 minutes on magnetic plate heater. After allowing the mixture to cool, it was filtered twice through Whatman filter paper to eliminate any remaining

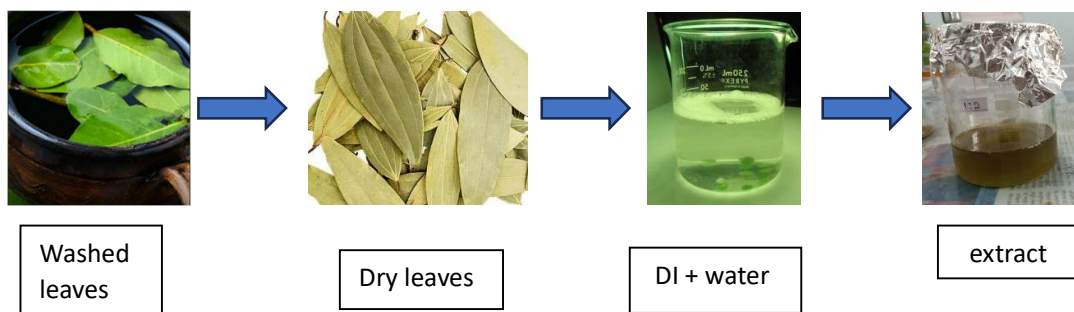


Fig5: Indian bay leave extract

particles, the filtrate that was left behind was the necessary extract. The extract was then stored at a cool place for further experimental.

For preparation of nickel nitrate solution, we took 2.32gram of nickel nitrate in 40ml of DI water and stirrer it for 10minute to prepare a nickel nitrate solution.

2.3 Synthesis of NiO nanoparticle

Sample 1 (yellow elder), was prepared by adding 40ml of flower extract in 40ml of nickel nitrate solution, then the sample was packed into the Teflon-lined stainless-steel autoclave and was annealed at 150⁰C for 3 hours. After this the autoclave was left to cool down at room temperature, the dark grey-black solution found and then the upper and lower part of the solution was collected into two separate beakers and these beakers were then heated in oven at 80⁰C till the solution is completely dry. Check this liquid solution in every 1-2hours until it gets dry in the form of powder. This powder was collected in crucible and calcinated at 600⁰C for 180 minutes in muffle furnace. The calcinated powder was collected into the sample bottle and undergo further analysis like UV, XRD, and SEM.

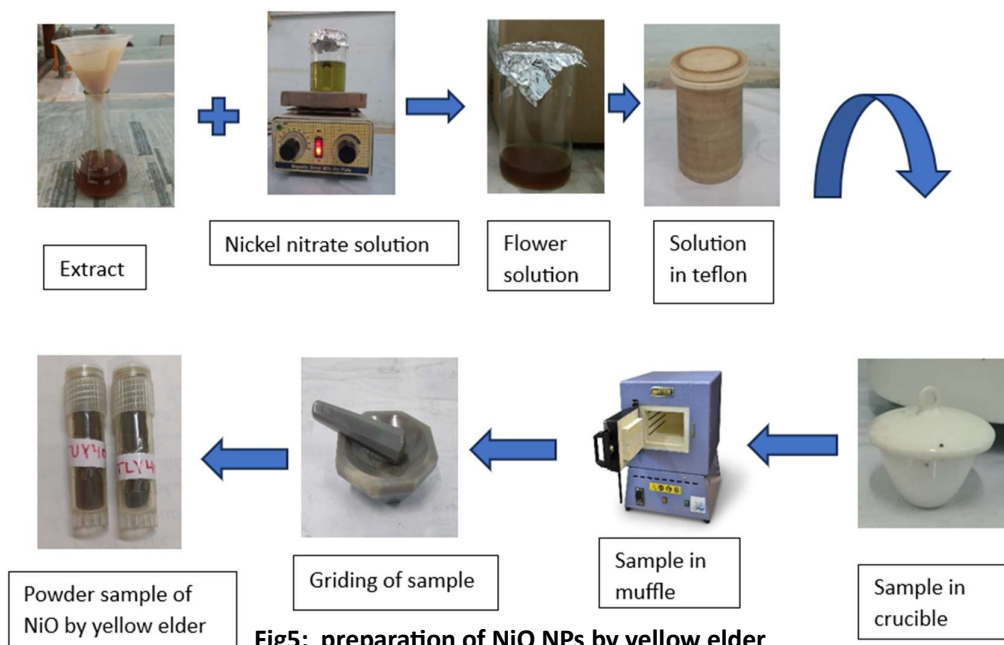


Fig5: preparation of NiO NPs by yellow elder

Sample 2 Indian bay leaf, was prepared by adding 20ml of leaf extract in 40ml of nickel nitrate solution, now this 60ml (20ml+40ml) solution will sonicate for 5minute to agitate the particles. Then the sample was packed into the Teflon-lined stainless-steel autoclave and was annealed at 150⁰C for 3 hours. After this the autoclave was left to cool down at room temperature, the dark green solution was found, then poured this solution in a beaker and left in the oven at 80⁰C till it got completely dry or came in powder form. The powder was collected into the sample bottle and underwent further analysis like UV, XRD, and SEM.

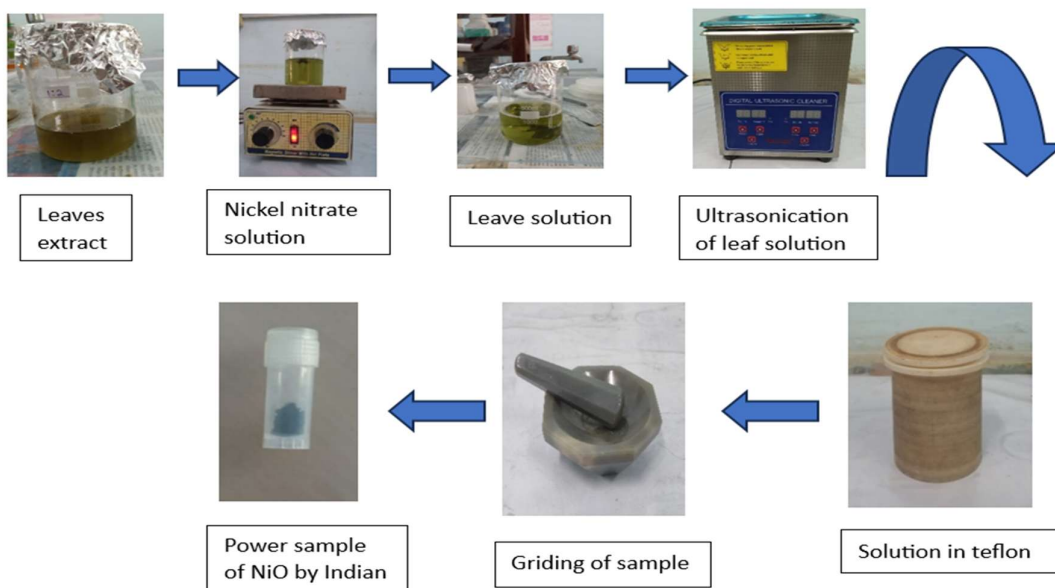


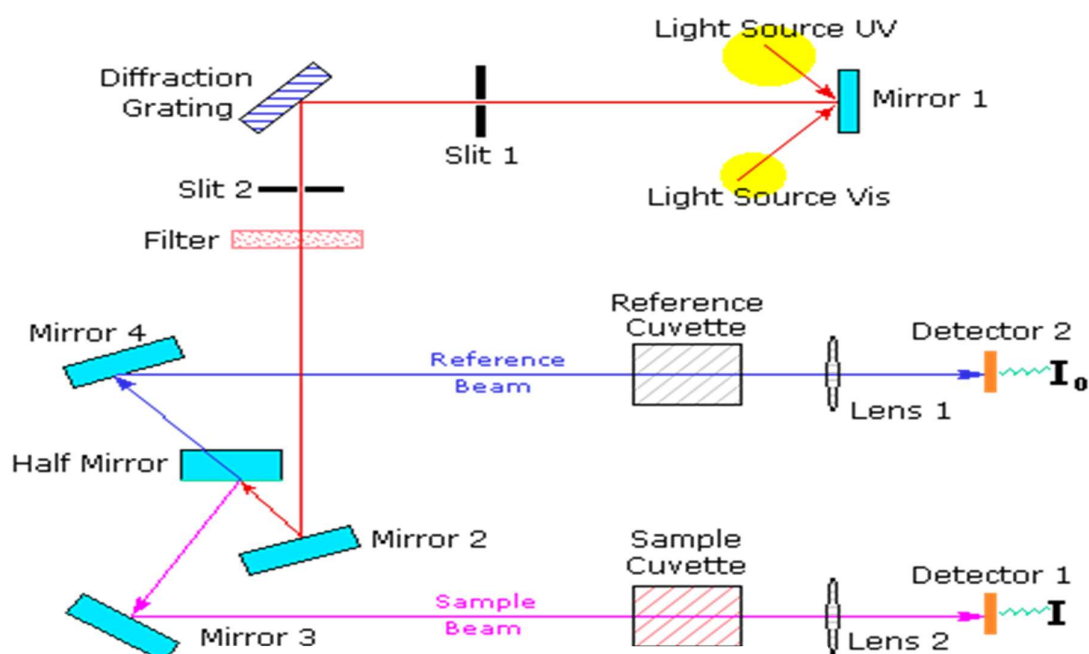
Fig6: preparation of NiO by Indian bay leaf

CHAPTER- 3 CHARACTERIZATION TECHNIQUE

3.1 UV-vis spectroscopy

Absorption and reflectance spectroscopy in the ultraviolet-visible spectral region are referred to as ultraviolet-visible spectroscopy or ultraviolet-visible spectrophotometry. Using ultraviolet-visible (UV-VIS) spectroscopy, an analytical technique, the quantity of the analyte can be determined based on how much light it absorbs. Measurements of the ultraviolet/visible area (UV-Vis) range in wavelength from around 200 nm to 800 nm. There are changes in the electrical energy levels of molecules as they absorb ultraviolet or visible light[29]. Characterization can benefit from an understanding of the optical and electrical characteristics of various materials, including liquids, films, powders, and monolithic solids.

An inexpensive, straightforward, adaptable, and non-destructive analytical method that works with a wide range of organic molecules and some inorganic species is UV-vis spectroscopy. UV-vis spectrophotometers detect the absorption or transmission of light through a material as a function of wavelength. UV-vis detectors are used in high performance liquid chromatography and ultra-high performance liquid chromatography to classify and measure the concentration of compounds in liquid streams. It combines these methods with mass spectrometry to enable the detection of all species.



The position of an electronic transition, whether it be rotational or vibrational, is primarily dictated by the change in electronic energy. The changes in the other states only result in neighbouring transitions and affect the electronic bands' vibrational and rotational structures. The intensity of these bands are related to the concentration C and the path length l by the Bouguer-Lambert-Beer law:

$$A = \log I_0/I = \log 1/T$$

where A is the absorbance or extinction, I_0 and I are the intensities of the monochromatic light before and after the sample, and T is transmittance respectively. The statement of the Beer-Lambert law can be written as follows: When a beam of monochromatic light is made incident on a solution that contains a substance that absorbs the monochromatic light, the rate at which the intensity of the beam decreases along the thickness of the solution is directly proportional to the concentration of the absorbing substance in the solution and is also directly proportional to the intensity of the incident monochromatic radiation.

3.2 X-ray diffraction

X-ray diffraction, phenomenon in which the atoms of a crystal, by virtue of their uniform spacing, cause an interference pattern of the waves present in an incident beam of x-ray. The atomic planes of the crystal act on the X-rays in exactly the same manner as does a uniformly ruled diffraction grating on a beam of light. A beam of X-rays contacts a crystal with an angle of incidence θ . It is reflected off the atoms of the crystal with the same angle θ . The X-rays reflect off atomic planes in the crystal that are a distance d apart[30].

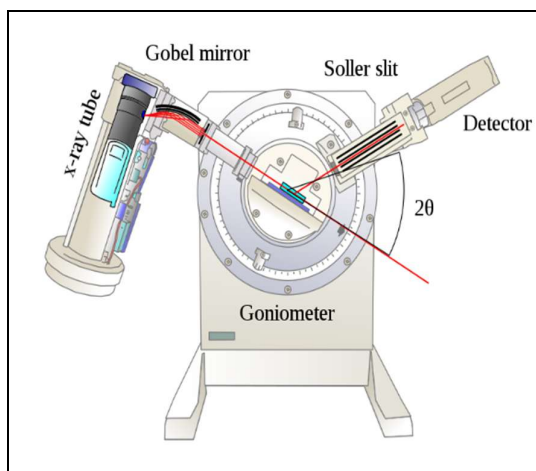


Fig8 (a): schematic diagram of XRD

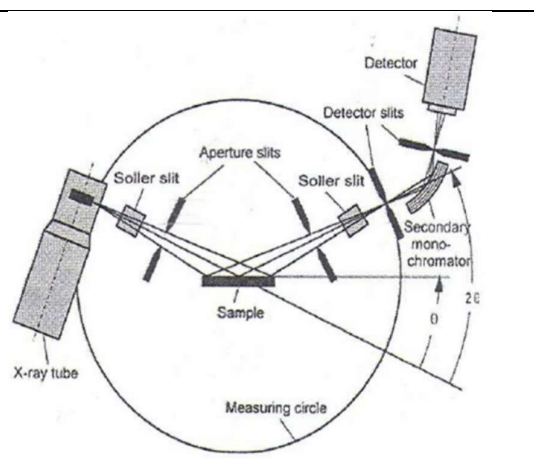


Fig8(b): Schematic representation of Bragg-Brentano diffractometer

The X-rays reflecting off two different planes must interfere constructively to form an interference pattern; otherwise, the X-rays would interfere destructively and form no pattern. To interfere constructively, the difference in path length between the beam reflecting off two atomic planes must be a whole number (n) of wavelengths (λ), or $n\lambda$. This leads to the Bragg law:

$$n\lambda = 2d \sin \theta$$

Within a cathode ray tube, X-rays are produced by burning a filament to produce electrons, adding a voltage to accelerate the electrons toward a target, and then saturating the target material with electrons. Typical X-ray spectra are generated when electrons have enough energy to remove the target material's inner shell electrons. These spectra are made up of many elements, the most prevalent of which are K_{α} and K_{β} . Part of K_{α} is made up of $K_{\alpha 1}$ and $K_{\alpha 2}$. $K_{\alpha 1}$ has double the strength of $K_{\alpha 2}$ and a somewhat shorter wavelength. The target material (Cu, Fe, Mo, and Cr) has particular wavelengths that are characteristic of it. In order to generate the monochromatic X-rays required for diffraction, filtering is necessary using foils or crystal monochrometers. Because $K_{\alpha 1}$ and $K_{\alpha 2}$ have similar wavelengths, a weighted average of the two is utilized. The most popular target material for single-crystal diffraction is copper, which has a CuK α radiation of $\lambda = 1.5418 \text{ \AA}$. The sample is exposed to these collimated X-rays. The intensity of the reflected X-rays is measured when the detector and sample are rotated. Constructive interference and the appearance of an intensity peak happen when the incident X-rays' geometry meets the requirements of Bragg's law and they impinge on the sample.

An output device, like a printer or computer monitor, receives the count rate generated by a detector that records an X-ray signal. The sample rotates at an angle θ along the path of the collimated X-ray beam in an X-ray diffractometer, and the X-ray detector, which is positioned on an arm to collect the diffracted X-rays, rotates at an angle of 2θ . A goniometer is the device that rotates and maintains the angle of the sample. Data are gathered at 2θ from 5° to 70° , preset angles in the X-ray scan, for typical powder patterns.

3.3 SEM (scanning electron microscopy)

One of the most useful tools for examining and analysing microstructure morphology and chemical composition characterizations is the scanning electron microscope (SEM). To comprehend the foundations of electron microscopy, one must be familiar with the fundamental ideas of light object. At an ideal viewing distance of 25 cm, the unassisted eye can distinguish

objects with a resolution of less than $\sim 0.1\text{mm}$, or roughly $1/60^\circ$ of visual angle. The maximum resolution achievable in optical microscopy is around $\sim 2,000 \text{ \AA}$ when using an optical lens to enlarge the viewing angle.

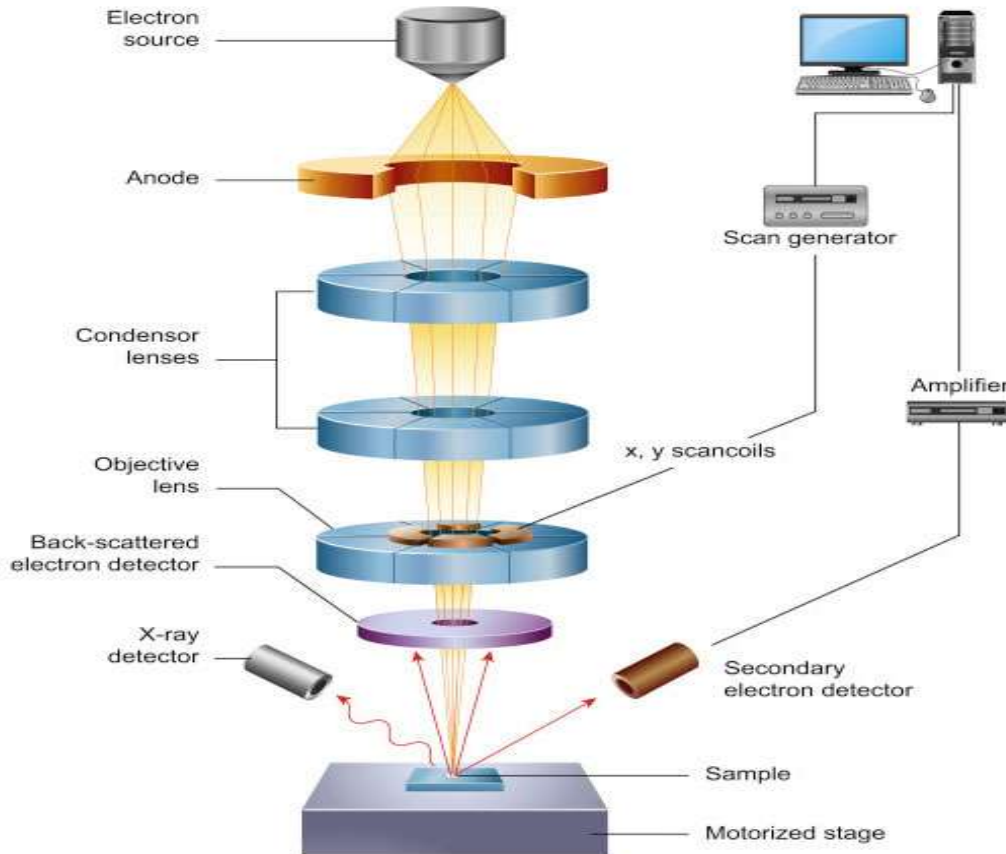


Fig9: Schematic diagram showing components of SEM and how they work

The SEM's ability to develop images depends on its ability to acquire signals from interactions between the material and the electron beam. Elastic and inelastic interactions are the two main groups into which these interactions can be separated. The incident electron is deflected by the specimen atomic nucleus or by electrons of a similar energy in the outer shell, which causes elastic scattering. This type of contact is characterized by a wide-angle directional change of the dispersed electron and a negligible energy loss during the collision. Backscattered electrons (BSE) are incident electrons that are elastically dispersed through an angle greater than 90° ; they produce a signal that is useful for imaging the material [31].

The main beam electron transfers a significant amount of energy to the sample atom by inelastic scattering, which is caused by a number of interactions between the incident electrons and the electrons and atoms of the sample. The specimen electrons' individual or collective excitation as well as the electron's binding energy to the atom determine how much energy is lost.

Therefore, during the ionization of specimen atoms, the specimen electrons are excited, resulting in the production of secondary electrons (SE), which are often described as having energies of less than 50 eV and can be utilized to scan or examine the sample.

CHAPTER-4 RESULT AND ANALYSIS

4.1 UV-Vis Spectral Study

Uv-vis spectroscopy of acquired NiO was performed by taking DI water as reference and the comparison of absorption of light between NiO and DI water was detected and the output was displayed on the computer screen graphs were plotted as absorbance Vs wavelength. The absorption peaks of TLY401 as, centred about were found 238nm and 323nm, *Deshpande, M. P., et al.* (2016) [32], UV-Vis spectroscopy of NiO solution as shown in fig10(a). The sample of solution show max absorption at 4.83eV and 3.76 eV as shown in fig10(b).

The absorption peaks of Nigreen, centred about were found 237nm, 284nm, 381nm *Sabouri, Zahra, et al* (2021)[33], in UV-Vis spectroscopy of NiO solution shown in fig11(a). These peaks were present due to extract react with Nickel II Nitrate. The sample of NiO solution show max absorption at , 4.00eV and 5.05eV as shown in fig11(b). The band gap of these nanoparticles can be calculated by using these direct formulae.

$$\alpha h\nu = A(h\nu - E_g)^m$$

Where,

α = absorption coefficient

$h\nu$ = energy of incident photo

A = energy constant

m = to be $\frac{1}{2}$ indirect allowed

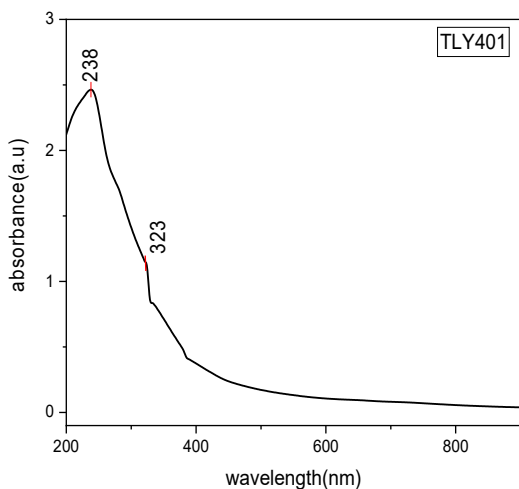


Fig10(a): Uv-Vis of TLY401

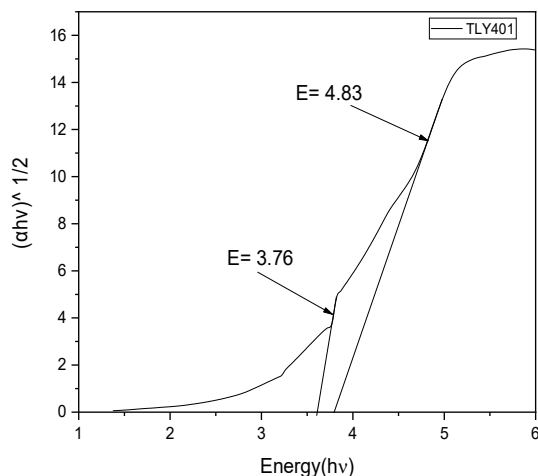


Fig10(b): energy band graph of TLY401

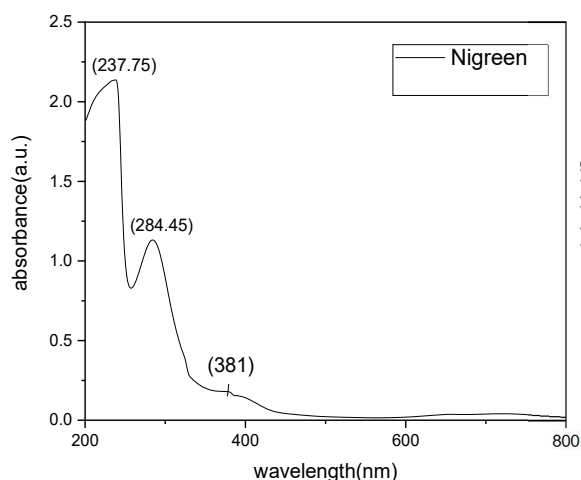


Fig11(a): Uv-Vis of Nigreen

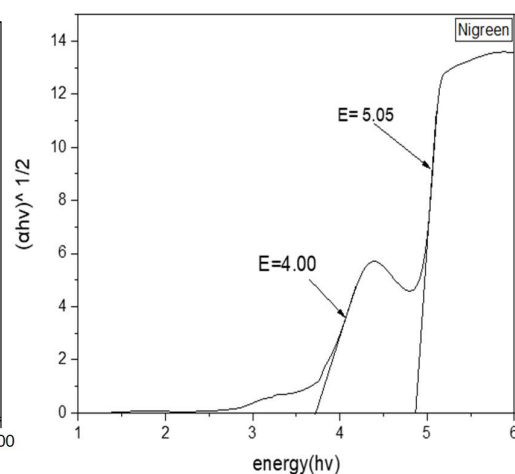


Fig11(b): energy band gap of Nigreen

4.2 X-ray diffraction

The structural properties of prepared NiO nanoparticles were studied under x-ray diffraction taking 2θ from 0-90 degree. Graph were plotted between 2θ values on X -Axis and intensity on y axis. The plotted XRD graph of TUY401 before calcination is as shown in fig12 shows no peaks on the other hand the graph of same sample (TLY401) showing peaks values after calcination as shown in fig13. Whereas the other sample i.e. Nigreen was showing peaks without calcination.

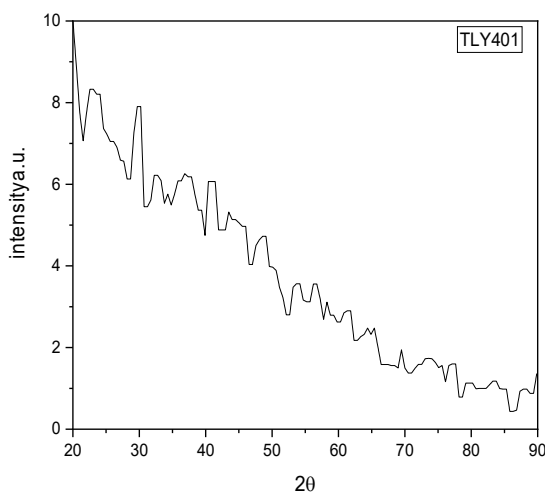


Fig12: XRD graph of TLY401 before calcination temperature

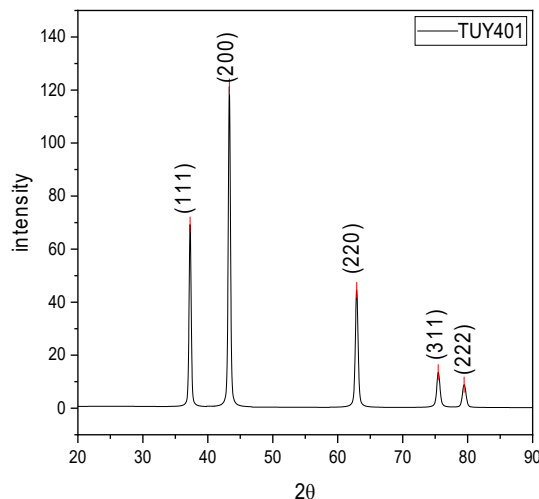


Fig13: XRD graph of TUY401 after calcination temperature

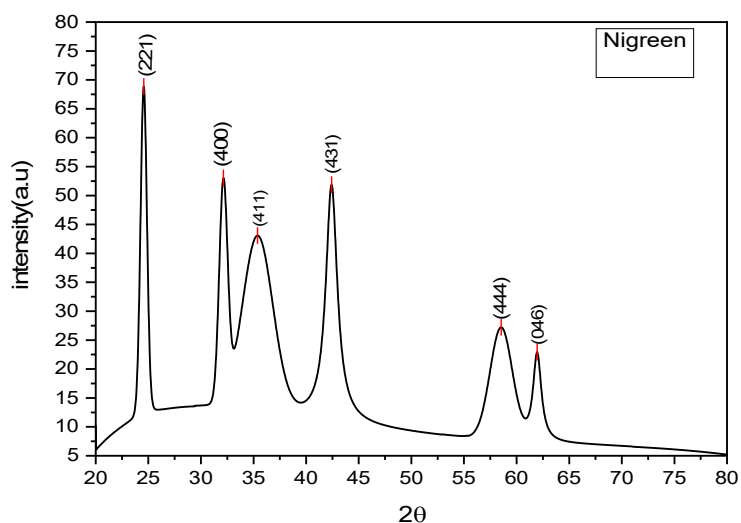


Fig14: XRD graph of Nigreen without calcination temperature

The peak values of TUY401 were found to be 37.29, 43.32, 62.91, 75.45, and 79.50, which were indexed to (111), (200), (220), (311), (222), of the face centered cubic unit cell, with reference to literature report (**JCPDS No: 01-071-1179**) The intensity was (200) peak was the highest followed by the other peaks. The average crystalline sizes estimated from XRD data using Scherrer equation were **18.67nm** for nano powder.

For the Nigreen sample the peak values were 24.57, 33.13, 35.37, 42.42, 58.53, 61.95 which were indexed to (221), (400), (411), (431), (444), (046) of the face centered cubic unit cell, with reference to literature report (**JCPDS No: 01-073-2113**) The intensity was (221) peak was the highest followed by the other peaks. The average crystalline sizes estimated from XRD data using Scherrer equation were **6.27nm** for nano powder.

4.3: SEM (SCANNING ELECTRONIC MICROSCOPY)

The structural features of the synthesised NiO-NPs were investigated through electron microscopy. The SEM images of tecoma stan(TLY401) reveal the layer type structure as shown in (fig16) and the size of crystal is **18.67nm**. The surface appears to be composed of various agglomerates and fine particles, with significant variation in size and shape. The detailed texture observed suggests the presence of both crystalline and amorphous regions. On the other hand SEM image of Indian bay leaf (Nigreen) shows spherical like structure and size of the aggregation is estimated to be **6.27nm**.

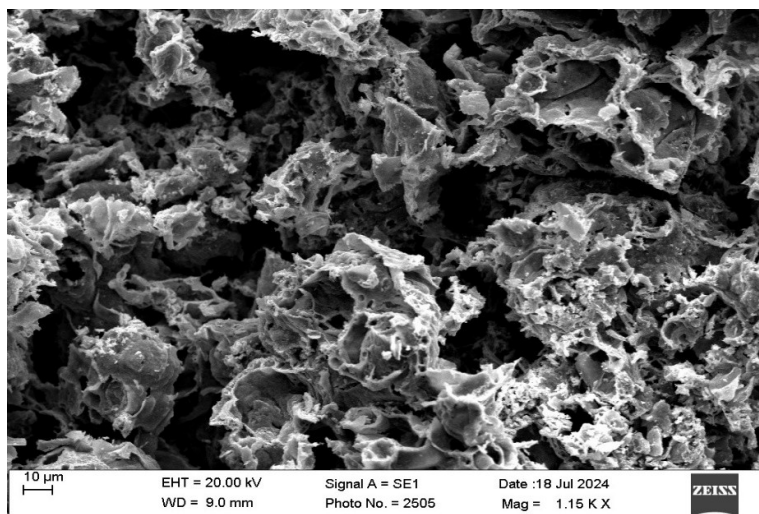


Fig16: SEM image of nano-nickel oxide of flower sample were calcinated at 600°C

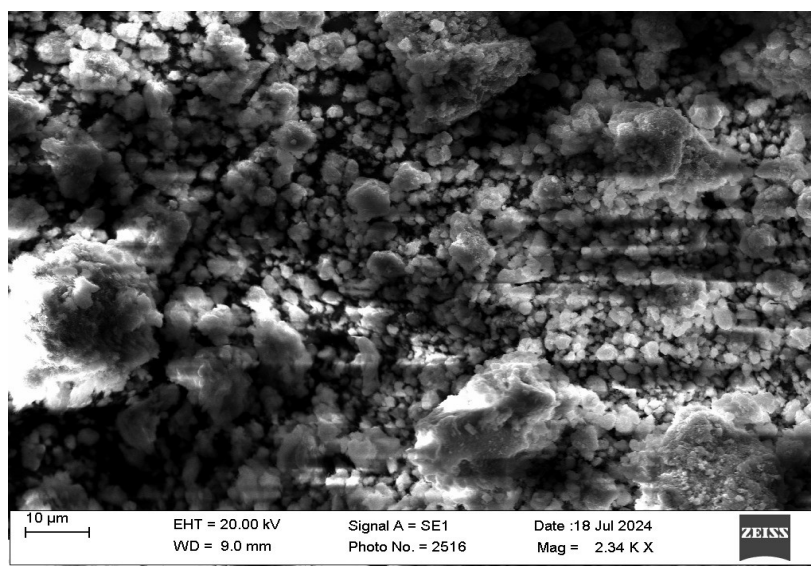


Fig15: SEM image of nano-nickel oxide of leaves sample devoid of calcination

CONCLUSIONS

In conclusion, selection of plant is very crucial in preparation of nanoparticles by green approach. Crystalline NiO nanoparticles are obtained from leaves extract only without post heating treatment. Formation of crystalline materials is seen after calcination at 600°C if yellow elder flower extract is used. Significant difference is also noticed in morphology in the NiO nanoparticles prepared by the extracts. More regular and allogerated shapes are seen by using leaves extract. However, layered agglomerated structure are seen by using flower extract and post synthesis heat treatment.

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